

# CS 342: Computer Vision and Pattern Recognition

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# Shanti Mantra

## Lyrics in Sanskrit

## Meaning in English

ॐ सह नाववतु।

Om, May we all be protected

सह नौ भुनक्तु।

May we all be nourished

सह वीर्यं करवावहै।

May we work together with great energy

तेजस्यि नावधीतमस्तु

May our intellect be sharpened (may our study be effective)

मा विद्विषावहै।

Let there be no Animosity amongst us

ॐ शान्तिः शान्तिः शान्तिः ॥

Om, peace (in me), peace (in nature), peace (in divine forces)



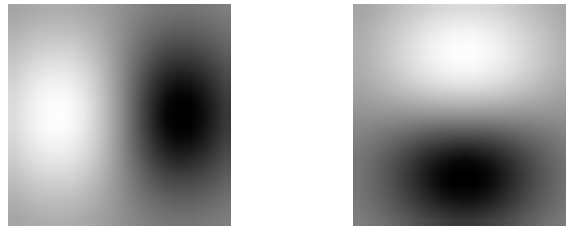
# Recall: image filtering

- Compute a function of the local neighborhood at each pixel in the image
  - Function specified by a “filter” or mask saying how to combine values from neighbors.
- Uses of filtering:
  - Enhance an image (denoise, resize, etc)
  - Extract information (texture, edges, etc)
  - Detect patterns (template matching)

# Template matching

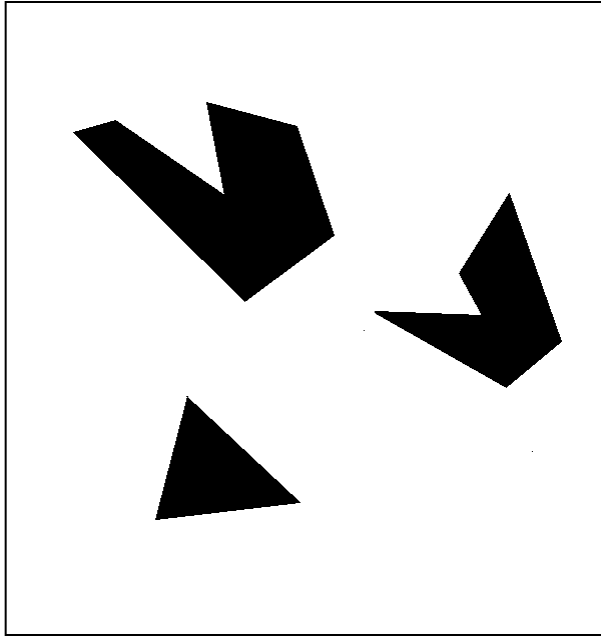
- Filters as **templates**:

Note that filters look like the effects they are intended to find --- “matched filters”

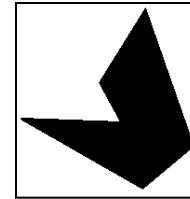


- Use normalized cross-correlation score to find a given pattern (template) in the image.
- Normalization needed to control for relative brightnesses.

# Template matching



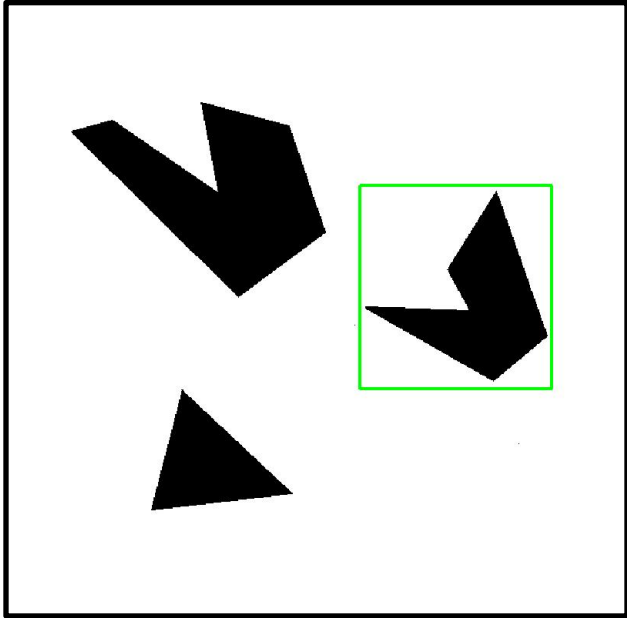
**Scene**



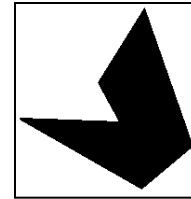
**Template (mask)**

**A toy example**

# Template matching

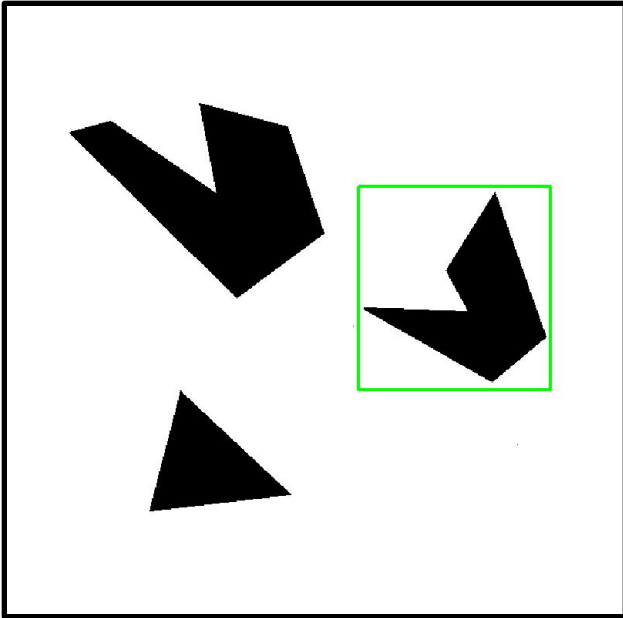


**Detected template**

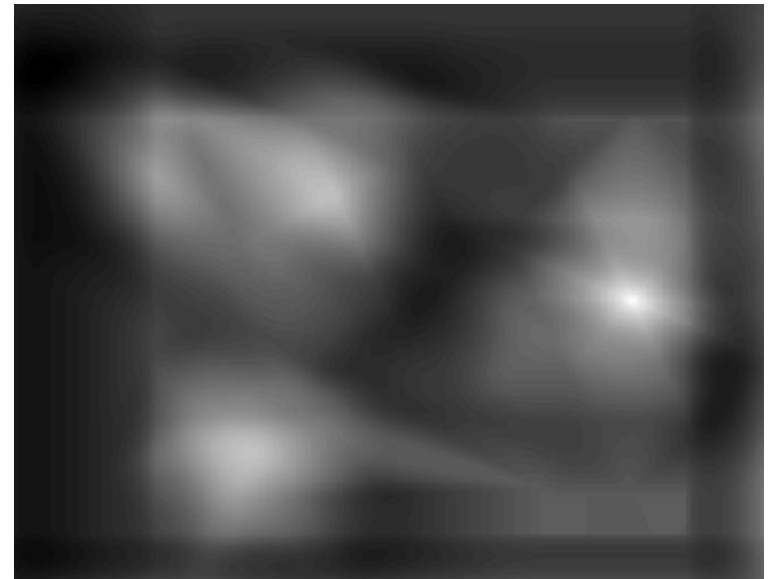


**Template**

# Template matching



**Detected template**



**Correlation map**

# Where's Waldo?



**Scene**



**Template**



# Where's Waldo?



**Detected template**



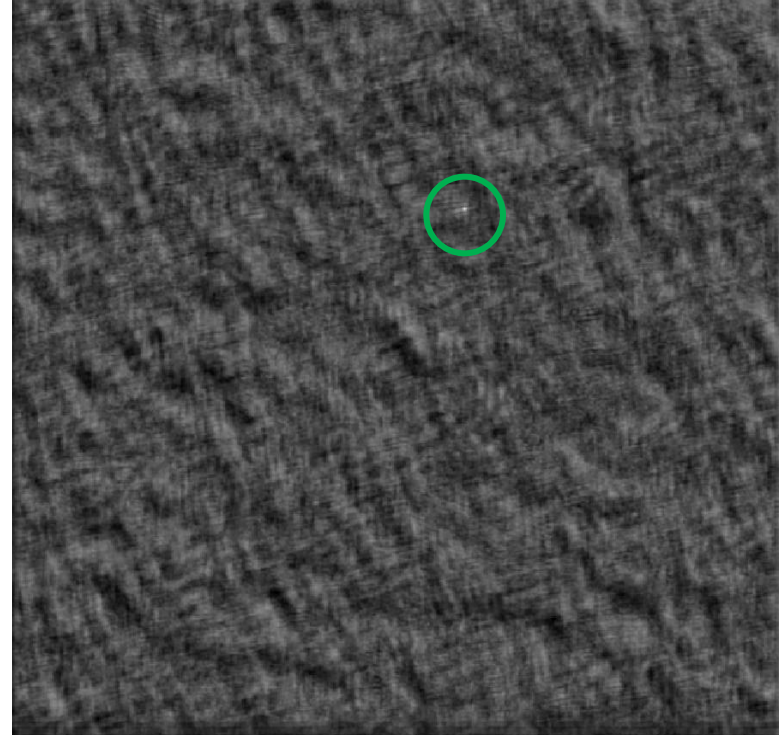
**Template**



# Where's Waldo?



**Detected template**



**Correlation map**

# Template matching



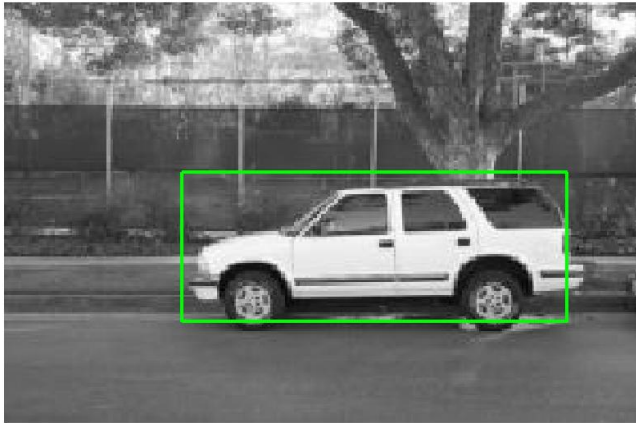
**Scene**



**Template**

What if the template is not identical to some subimage in the scene?

# Template matching



**Detected template**



**Template**

Match can be meaningful, if scale, orientation, and general appearance is right.

How to find at any scale?



# OpenCV Template Matching

```
res = cv2.matchTemplate(img,template,method)  
min_val, max_val, min_loc, max_loc = cv2.minMaxLoc(res)
```

# method

| Enumerator       |                                                                                                                                                                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TM_SQDIFF        | $R(x, y) = \sum_{x', y'} (T(x', y') - I(x + x', y + y'))^2$                                                                                                                                                                                                |
| TM_SQDIFF_NORMED | $R(x, y) = \frac{\sum_{x', y'} (T(x', y') - I(x + x', y + y'))^2}{\sqrt{\sum_{x', y'} T(x', y')^2 \cdot \sum_{x', y'} I(x + x', y + y')^2}}$                                                                                                               |
| TM_CCORR         | $R(x, y) = \sum_{x', y'} (T(x', y') \cdot I(x + x', y + y'))$                                                                                                                                                                                              |
| TM_CCORR_NORMED  | $R(x, y) = \frac{\sum_{x', y'} (T(x', y') \cdot I(x + x', y + y'))}{\sqrt{\sum_{x', y'} T(x', y')^2 \cdot \sum_{x', y'} I(x + x', y + y')^2}}$                                                                                                             |
| TM_CCORR_NORMED  | $R(x, y) = \sum_{x', y'} (T'(x', y') \cdot I'(x + x', y + y'))$ <p>where</p> $T'(x', y') = T(x', y') - 1/(w \cdot h) \cdot \sum_{x'', y''} T(x'', y'')$ $I'(x + x', y + y') = I(x + x', y + y') - 1/(w \cdot h) \cdot \sum_{x'', y''} I(x + x'', y + y'')$ |
| TM_CCORR_NORMED  | $R(x, y) = \frac{\sum_{x', y'} (T'(x', y') \cdot I'(x + x', y + y'))}{\sqrt{\sum_{x', y'} T'(x', y')^2 \cdot \sum_{x', y'} I'(x + x', y + y')^2}}$                                                                                                         |

# Recap: Mask properties

- Smoothing

- Values positive
- Sum to 1  $\rightarrow$  constant regions same as input
- Amount of smoothing proportional to mask size
- Remove “high-frequency” components; “low-pass” filter

- Derivatives

- Opposite signs used to get high response in regions of high contrast
- Sum to 0  $\rightarrow$  no response in constant regions
- High absolute value at points of high contrast

- Filters act as templates

- Highest response for regions that “look the most like the filter”
- Dot product as correlation



# Summary

- Image gradients
- Seam carving – gradients as “energy”
- Gradients  $\rightarrow$  edges and contours
- Template matching
  - Image patch as a filter

# Coming up

- Project 1 is going to be out soon.